

Chang He

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Research interests: Bioinformatics/Systems biology/Transcriptomics

EDUCATION

Tsinghua University, Beijing

Aug 2022 - Jun 2026 (expected)

Bachelor of Science: Biological Sciences, Statistics (minor) GPA 3.84/4.00

Related Courses:

- Biology-related: *Biochemistry, Molecular Biology (honor), Genetics, Cell Biology, Systems Biology, Physiology*
- Statistics-related: *Statistical Inference, Multivariate Statistical Analysis, Introduction to Bayesian Statistics, Statistical Computing and Software, Elementary Probability Theory, Linear Regression Analysis, Programming Fundamentals*

RESEARCH EXPERIENCES

A generalizable framework bridging cross-context transcriptomes for cancer therapy prediction

Advisor: Siwei Chen & Junhan Zhao

University of Chicago 2025.8 - present

- Developed ASPECT, a LLM-based foundation framework tokenizing RNA-seq profiles into “cell sentences” to map cross-context transcriptome semantic alignment.
- Formulated a clinical “Benefit Score”, achieving an average AUC improvement of 0.35 for systemic cytotoxic agents.
- Extended the semantic alignment workflow to single-cell resolution to decipher extrinsic immune microenvironment dynamics.

Transcriptomic analysis of tRNA dynamics and codon-specific translation in the human brain

Advisor: Tao Pan

University of Chicago 2025.6 - 2025.8

- Expanded the tRNA-seq analysis pipeline by integrating ML, PCA, batch correction, and tRNA visualization.
- Analyzed and compared codon usage bias based on Ribo-seq, proteomics, and bulk mRNA-seq data.
- Analyzed tRNA dynamics, including abundance, modification, and fragmentation in Alzheimer’s and ALS patients.
- Concluded the preliminary results for a panel in grant application.

Development of novel metric of cell-free RNA fragmentation revealing RNA metabolism

Advisor: Zhi John Lu

Tsinghua University 2025.1 - 2025.10

- Designed a statistical model to calculate a novel metric and enhanced cancer classification.
- Examined the capability of the metric to mitigate the batch effect of cell-free RNA datasets.
- Discovered the biological mechanism behind the metric by mining both cell-free and cellular TCGA data.
- Drafted and submitted a manuscript as a co-first author.

Representation Learning of plasma cell-free RNA for systemic lupus erythematosus diagnosis

Advisor: Zhi John Lu

Tsinghua University 2024.8 - 2025.2, 2025.12 - 2026.6

- Developed a late-fusion ensemble model utilizing dual-track foundation models to capture under-characterized non-canonical RNAs, achieving a 0.10 average AUC improvement across 6 organ involvements.
- Validated that non-canonical RNA clades drive organ-specific predictions (e.g., Renal, Pulmonary) that are conventionally missed by traditional mRNA-centric analysis.
- Calculated cell-specific multi-omics features (mRNA, TE RNA, alternative splicing) via machine learning to identify key contributing cell types
- Built an online data portal (sledatportal.vercel.app) to host analytics on dsRNA.

Establishment of a scRNA-seq dataset reliability assessment system

Advisor: Bin Li

National Institute of Biological Sciences, Beijing 2023.12 - 2024.6

- Built a clustering strategy on scRNA-seq data to annotate cells, describing their typicalities.
- Discovered the system’s biological significance in diseases and cell development.
- Integrated and examined “center” cells to improve clustering and annotation performance.

ACADEMIC PAPERS

- **Chang H.**, Junhan Z.^{*}, Zheyi J., Chenwei T., Zechuan Z., Anbang L., Siwei C.^{*}. “A generalizable language model framework bridging cross-context transcriptomes for cancer therapy prediction” submitted.

- Kaicheng U[†], Sophia M. Z.[†], Ziyu Y., Jianwei Z., Zechuan Z., Shichao D., **Chang H.**, Lavonda L., Rui C., Zongxin Y., Junhan Z.^{*}, Siwei C.^{*}, and Xingcai Z.^{*}. "AI-Powered Medicinal Chemistry: From Molecular Design to Clinical Development" Chemical Society Reviews, accepted.
- Yilan B.[†], **Chang H.**[†], Yuhuan T., Pengfei B., and Zhi J. L.^{*}. "Leveraging Fragmentation Patterns of cell-free RNA to Mitigate Batch Effects for Cancer Classification" in manuscript.

AWARDS

China High School Biology Olympiad 1st Prize

National gold award

Best Report Award

For excellent reporting in *General Biology Fieldwork* course

Academic Excellence Scholarship

For exceptional academic performance (Top 10%)

Technology Innovation Excellence Scholarship

For outstanding results in scientific and technological practical activities (Top 10%)

China Zoological Society & Botanical Society of China

Aug 2021

School of Life Sciences, Tsinghua University

Sep 2023

Zhili College, Tsinghua University

Oct 2023

Tsinghua University

Oct 2025

ACADEMIC PRESENTATION

Cell-specific novel RNA features in lupus organ involvements

20-minute presentation for *Biological Research Training (honor)* course

Dec 2024

Profiling tRNA Dynamics in Alzheimer's in a Sex-Specific Manner

Oral presentation at the UChicago-THU Summer Research Seminar, University of Chicago.

Aug 2025

SKILLS

• Computational & Bioinformatics:

- **Bioinformatics Analysis:** Statistical modeling & Machine learning on high-dimensional omics data, LLM inference with RAG for biological text processing, Single-cell RNA-sequencing (scRNA-seq) analysis, Bulk RNA-sequencing (RNA-seq) pipeline development and execution.
- **Programming Languages:** R (extensive experience in bioconductor packages and data visualization); Python (Pandas, NumPy, Scikit-learn); Bash scripting; C (Foundational knowledge).
- **Tools & Version Control:** Git/GitHub, HPC/Linux Environment management.

• Technical & General:

- **Web Development:** Typescript/Javascript.

• Language Proficiency:

Chinese (Native); English (TOEFL 110); Portuguese (Intermediate).

REFERENCES

Prof. Zhi John Lu

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Scholar Profiles: Personal Page — Google Scholar — Lab Website

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